

Thermal Equilibrium - when objects have been in contact and the macroscopic coordinate (variables) have stopped changing.

→ involves an exchange of energy between two objects, or an object & its surroundings

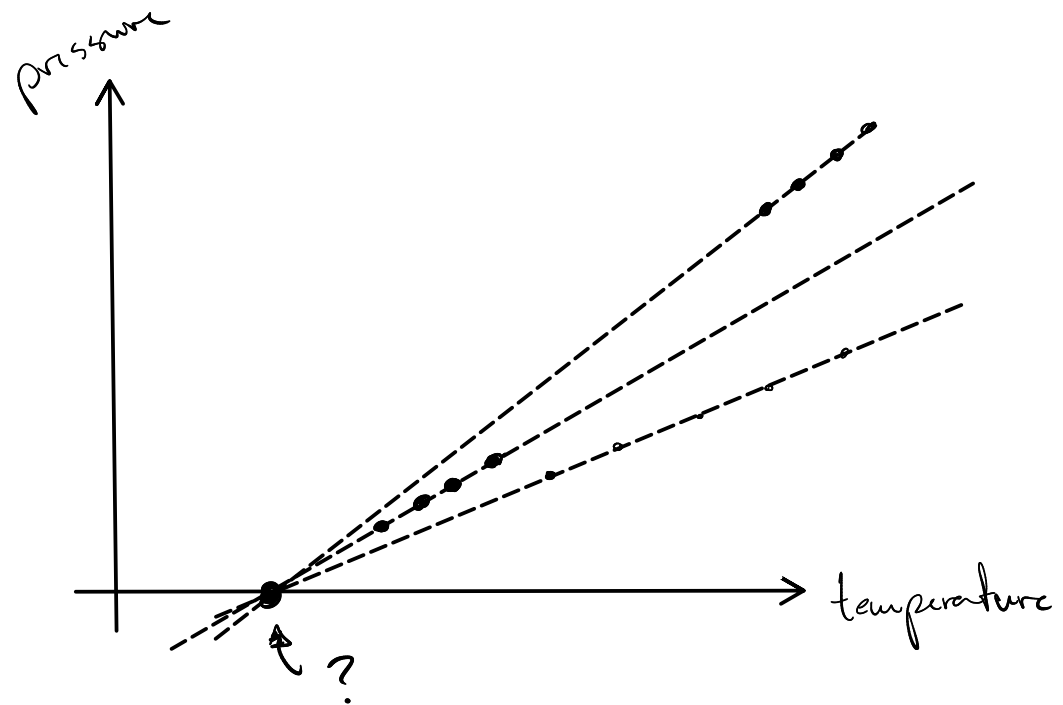
↳ volume

- pressure
- electrical resistance
- radiation / thermal emf

$$\begin{array}{ll} \text{I} & F_1 = 32^\circ\text{F} \quad F_2 = 212^\circ\text{F} \\ & C_1 = 0^\circ\text{C} \quad C_2 = 100^\circ\text{C} \end{array}$$

$$\hookrightarrow C = mF + b$$

$\downarrow \quad \quad \downarrow$
 $? \quad \quad ?$



Equipartition Theorem

$$U_{\text{thermal}} = N \cdot f \cdot \frac{1}{2} k_B T$$

$$f = 3 \leftarrow \text{monatomic gas}$$

$$f = 5 \leftarrow \text{diatomic gas}$$

$$N k_B = n R$$

$$U = n \cdot f \cdot \frac{1}{2} R T$$

$$\hookrightarrow U = \frac{3}{2} n R T \quad (\text{monatomic})$$

$$\hookrightarrow U = \frac{5}{2} n R T \quad (\text{diatomic})$$

Kinetic Energy

$$E_k = \frac{1}{2} m v^2$$

$$\vec{v} = v_x \hat{x} + v_y \hat{y} + v_z \hat{z}$$

$$E_k = \frac{1}{2} m \vec{v} \cdot \vec{v}$$

$$E_k = \frac{1}{2} m (v_x^2 + v_y^2 + v_z^2)$$

Spring Potential

$$U_{sx} = \frac{1}{2} k_x x^2$$

$$U_{sy} = \frac{1}{2} k_y y^2$$

$$U_{sz} = \frac{1}{2} k_z z^2$$

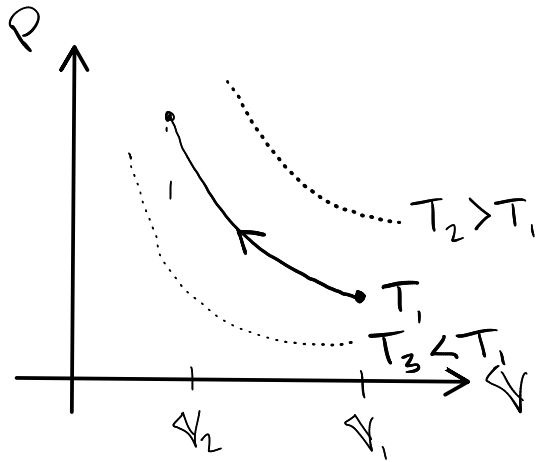
Rotation

$$K_{\text{rot}} = \frac{1}{2} I \omega^2$$

Isobaric (constant pressure) + Isochoric (constant volume), we have discussed in worksheet (week 3)

Isothermal (constant temperature)

↳ compression



$$PV = Nk_B T$$

constant!

$$P = \frac{Nk_B T}{V} \leftrightarrow \text{hyperbola}$$

$$W = - \int P dV$$

$$= -Nk_B T \int_{V_1}^{V_2} \frac{1}{V} dV$$

$$\boxed{W = -Nk_B T \ln\left(\frac{V_2}{V_1}\right)} \rightarrow W = Nk_B T \ln\left(\frac{V_1}{V_2}\right)$$

so what about the heat?

first law $\Delta U = Q + W$

$$\Delta U = \frac{f}{2} Nk_B \Delta T$$

$$Q = -W$$

$$Q = Nk_B T \ln\left(\frac{V_2}{V_1}\right)$$

Adiabatic Process

↳ compression

(isentropic) (no heat added or removed)

$$\text{first law: } \Delta U = \cancel{Q} + W$$

$$dU = \cancel{dQ} + dW$$

$$dU = dW$$

$$\int \frac{f}{2} N k_B dT = - \int p dV$$

$p = \frac{N k_B T}{V}$

$$\int \frac{f}{2} \cancel{N k_B} dT = - \frac{\cancel{N k_B T}}{V} dV$$

$$\int \frac{f}{2} dT = - \int \frac{T dV}{V}$$

$$\frac{f}{2} \int_{T_1}^{T_2} \frac{dT}{T} = - \int_{V_1}^{V_2} \frac{dV}{V}$$

$$\frac{f}{2} \ln\left(\frac{T_2}{T_1}\right) = - \ln\left(\frac{V_2}{V_1}\right)$$

$$\ln\left[\left(\frac{T_2}{T_1}\right)^{\frac{f}{2}}\right] = \ln\left(\frac{V_1}{V_2}\right)$$

$$PV = N k_B T$$
$$p = \frac{N k_B T}{V}$$
$$W = \int p dV$$

$$y = - \frac{dy}{dx}$$

↳ $y = \text{~~~~~}$

$$\boxed{V \propto \frac{1}{T_{f/2}}}$$



$$\left(\frac{T_2}{T_1} \right)^{\frac{f}{2}} = \left(\frac{V_1}{V_2} \right)$$